

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 5 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|------------------------------|---|
| 1. Index _____ | (A) Everything written underneath the radical symbol; when the index is even, it may not be negative. |
| 2. Radicand _____ | (B) The operation that feeds one function's output into another function's input, written $(f \circ g)(x)$. |
| 3. Principal root _____ | (C) The small number in the crook of the radical symbol that says which power is being undone. |
| 4. Conjugate _____ | (D) The point (h, k) that positions a transformed radical graph — an endpoint when the index is even, a turning point when it is odd. |
| 5. Extraneous solution _____ | (E) A candidate that satisfies the powered-up equation but fails when it is tested in the original. |
| 6. Composition _____ | (F) The non-negative value the radical symbol by itself returns; a $\pm$ must be written in by hand. |
| 7. Inverse function _____ | (G) The function that reverses another's inputs and outputs; its graph is the mirror image over the line $y = x$. |
| 8. Anchor point _____ | (H) A binomial with the same two terms but the opposite middle sign; multiplying by it turns an irrational denominator rational. |

Part B — Multiple Choice (12 pts)

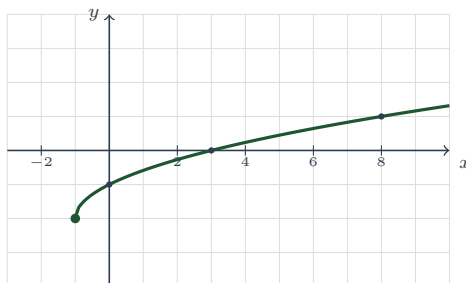
Circle the best answer.

- Written with a rational exponent, $\sqrt[3]{x^5}$ is (A) $x^{5/3}$ (B) $x^{3/5}$ (C) x^{15} (D) $x^{-3/5}$
- The domain of $f(x) = \sqrt{7-2x}$ is (A) $x \geq \frac{7}{2}$ (B) $x \geq -\frac{7}{2}$ (C) $x \leq \frac{7}{2}$ (D) all real numbers
- The graph of $y = \sqrt{3x-12} + 5$ begins at the point (A) (12, 5) (B) (4, -5) (C) (-4, 5) (D) (4, 5)
- In simplest form, $\sqrt{75} + \sqrt{12}$ is (A) $\sqrt{87}$ (B) $7\sqrt{3}$ (C) $7\sqrt{6}$ (D) $10\sqrt{3}$
- Which equation has *no real solution*?
(A) $\sqrt[3]{x+1} = -4$ (B) $\sqrt{x-1} = 7$ (C) $\sqrt{x+6} = -2$ (D) $\sqrt{x} = 0$
- If $f(x) = \sqrt{x}$ and $g(x) = x + 9$, then $(f \circ g)(x) =$ (A) $\sqrt{x} + 9$ (B) $\sqrt{9x}$ (C) $x\sqrt{x} + 9\sqrt{x}$ (D) $\sqrt{x+9}$

Part C — Short Answer & Computation (40 pts)

Show your work. Assume every variable is non-negative unless told otherwise.

1. **Read the graph.** The radical function f is graphed below; each gridline is one unit and the solid dot is the endpoint.



- (a) Coordinates of the endpoint: _____
- (b) Domain (interval notation): _____; range: _____
- (c) x -intercept: _____; y -intercept: _____
- (d) Interval(s) where f is increasing: _____
- (e) Does f have an absolute maximum or an absolute minimum? Give it and say where it occurs.

- (f) End behavior: as $x \rightarrow \infty$, $f(x) \rightarrow$ _____; as $x \rightarrow -\infty$, _____
- (g) Interval(s) where $f(x) < 0$: _____

2. **Radicals and rational exponents.** Rewrite or evaluate. Leave no negative exponents.

- (a) Write $\sqrt[4]{x^3}$ with a rational exponent. _____
- (b) Write $y^{5/2}$ as a radical. _____
- (c) $27^{2/3} =$ _____ (d) $16^{-1/2} =$ _____
- (e) Write $x^{1/2} \cdot x^{1/3}$ as a single radical. _____

3. **Simplify. Add or subtract where you can.**

(a) $\sqrt{72}$ (b) $\sqrt[3]{54}$

(c) $\sqrt{20x^3}$ (d) $5\sqrt{12} - \sqrt{27}$

4. **Multiply, then rationalize.** Every answer must be in simplest form.

(a) $\sqrt{6} \cdot \sqrt{15}$ (b) $(2 + \sqrt{5})(2 - \sqrt{5})$

(c) $\frac{6}{\sqrt{3}}$ (d) $\frac{4}{3 - \sqrt{5}}$

5. **Transformations.** Let $g(x) = -2\sqrt{x+3} + 4$.

- (a) List every transformation that takes $y = \sqrt{x}$ to g . _____
 - (b) Anchor point: _____
 - (c) Domain: _____; range: _____
 - (d) Is the anchor an absolute maximum or an absolute minimum? _____
 - (e) A different graph of the form $y = \sqrt{x-h} + k$ begins at $(5, -2)$. Write its equation. _____
 - (f) For $p(x) = \sqrt[3]{x-1} + 2$, give the anchor point _____ and the domain _____. In one sentence, say why that domain is not like g 's.
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6. **Solve. Test every candidate in the original equation.**

(a) $\sqrt{2x+3} = x$

(b) $\sqrt[3]{x-4} = -2$

(c) $\sqrt{3x-2} + 4 = 1$

(d) $x^{2/3} = 25$

7. **Composition.** Let $f(x) = \sqrt{x}$ and $g(x) = x - 6$.

- (a) $(f \circ g)(x) =$ _____, domain: _____
 - (b) $(g \circ f)(x) =$ _____, domain: _____
 - (c) Are these the same function? Answer using *both* the rule and the domain.
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(d) $(f \circ g)(10) =$ _____

(e) For $p(x) = 2x + 1$ and $q(x) = x^2 - 3$, find $(p \circ q)(-2)$. _____

8. **Inverses.**

- (a) $f(x) = 4x - 7$. Find $f^{-1}(x)$. _____
- (b) Verify by composition — *both* orders — that your answer really is the inverse.

(c) $g(x) = \sqrt{x+2}$. Find $g^{-1}(x)$ and state its domain.

(d) The point $(7, 3)$ lies on the graph of g . What point must lie on the graph of g^{-1} ? _____

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The full analysis.** Consider $f(x) = \sqrt{x-1} + 2$. **(a)** Give the anchor point, the domain, and the range, both in interval notation. **(b)** Find the x -intercept and the y -intercept, or explain why one does not exist. **(c)** List the transformations that take $y = \sqrt{x}$ to f . **(d)** Find $f^{-1}(x)$ and state its domain, saying exactly where that restriction comes from. **(e)** Justify with composition — both orders — that f and f^{-1} are inverses. **(f)** A classmate writes $y = (x-2)^2 + 1$ with *no* restriction and calls it the inverse. Explain why that is not the inverse of f .

2. **Modeling.** Ignoring friction, a roller-coaster car that starts from rest and drops h feet reaches the bottom at a speed of

$$v(h) = 8\sqrt{h} \text{ feet per second.}$$

- (a)** State the domain that makes sense here, and connect it to the rule about what an even index will accept.
- (b)** Find $v(9)$ and $v(36)$, with units. What does quadrupling the drop do to the speed, and which feature of $\sqrt{\cdot}$ is responsible?
- (c)** A car reaches 56 feet per second at the bottom. Find the height of the drop algebraically, and check your answer.
- (d)** Write the inverse, h as a function of v , and say what it tells an engineer that $v(h)$ does not.
- (e)** Part (c) required squaring both sides. Explain why there is no danger of an extraneous solution in this problem.